Slide Set #5

CLRM: sample properties of OLS

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Notes

By finite-sample we mean $n < +\infty$, that is finite sample size

Note: the terminology "*finite sample*" means a sample size n that is not "*too large*". When n is "*large enough*", the asymptotic regime takes over, but this will be the object of the second part of the course.

We will investigate properties of both $\boldsymbol{b}$ and s^2

A1–A4 are assumptions from previous slide sets

Notes

Finite sample properties of $\boldsymbol{b}$

Proposition: unbiasedness of $\boldsymbol{b}$

Assume A1, A2 and A3. Then $\mathrm{E}[\boldsymbol{b} \,|\, \boldsymbol{X}] = \boldsymbol{\beta}$, that is $\boldsymbol{b}$ is an unbiased estimator of $\boldsymbol{\beta}$ conditional on $\boldsymbol{X}$.

Proof. First note that by the linearity of expectations

$$\mathrm{E}[\boldsymbol{b} \,|\, \boldsymbol{X}] = \boldsymbol{\beta} \quad \text{whenever} \quad \mathrm{E}[\boldsymbol{b} - \boldsymbol{\beta} \,|\, \boldsymbol{X}] = \boldsymbol{0}$$

Recall the decomposition of the estimation error

$$\boldsymbol{b} - \boldsymbol{\beta} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\varepsilon}$$

then

$$\mathrm{E}[\boldsymbol{b} - \boldsymbol{\beta} \,|\, \boldsymbol{X}] = \mathrm{E}\left[(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}\right]$$

pull out what's known and obtain

$$\mathrm{E}[\boldsymbol{b} - \boldsymbol{\beta} \,|\, \boldsymbol{X}] = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\,\mathrm{E}\left[\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}\right] = \boldsymbol{0}$$

■

Notes

Note that this is a conditional unbiasedness statement, and it is stronger than the more traditional unconditional statement.

This is stronger because

$$\mathrm{E}[\boldsymbol{b} \,|\, \boldsymbol{X}] = \boldsymbol{\beta} \implies \mathrm{E}[\mathrm{E}[\boldsymbol{b} \,|\, \boldsymbol{X}]] = \mathrm{E}[\boldsymbol{b}] = \boldsymbol{\beta}$$

Notes

Proposition: variance-covariance of $\boldsymbol{b}$

Assume A1, A2 and A3 and A4. Then the followings hold:

(a) $\mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}] = \sigma^2 (\boldsymbol{X}'\boldsymbol{X})^{-1}$

(b) (Gauss-Markov theorem) Let $\boldsymbol{b}_0$ be any estimator of $\boldsymbol{\beta}$ that is linear in the $\boldsymbol{y}$ and unbiased. Then

$$\mathrm{Var}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] \succeq \mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}]$$

(c) $\mathrm{Cov}[\boldsymbol{b}, \boldsymbol{e} \,|\, \boldsymbol{X}] = \boldsymbol{0}$

Recall that $\succeq$ means bigger or equal for certain matrices. Since of **(b)** the OLS estimator is also called BLUE = *Best Linear Unbiased Estimator*. Best here is in terms of mean square error (MSE). Since $\boldsymbol{b}$ is unbiased, its efficiency is equal to its variance.

Notes

Proof of part (a). Since $\boldsymbol{\beta}$ is not a random variable, then its variance is zero, then

$$\begin{aligned}
\mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}] &= \mathrm{Var}[\boldsymbol{b} - \boldsymbol{\beta} \,|\, \boldsymbol{X}] \\
&= \mathrm{Var}[(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}] \\
&= \mathrm{Var}[\boldsymbol{A}\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}], \quad \text{for } \boldsymbol{A} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' \\
&= \boldsymbol{A}\mathrm{Var}[\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}]\boldsymbol{A}' \\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' \,\mathrm{Var}[\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}]\, \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} \qquad (*) \\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' \,(\sigma^2)\boldsymbol{I}_n\, \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} \\
&= \sigma^2 (\boldsymbol{X}'\boldsymbol{X})^{-1}\,(\boldsymbol{X}'\boldsymbol{X})\,(\boldsymbol{X}'\boldsymbol{X})^{-1} \\
&= \sigma^2 (\boldsymbol{X}'\boldsymbol{X})^{-1}
\end{aligned}$$

(*) this is because for a matrix $\boldsymbol{B}$, we have that $(\boldsymbol{B}')^{-1} = (\boldsymbol{B}^{-1})'$ ■

Notes

Proof of part (b). First note that OLS is linear map of $\boldsymbol{y}$. In fact, define $\boldsymbol{A} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$, then $\boldsymbol{b} = \boldsymbol{A}\boldsymbol{y}$.

By assumption there exists an alternative linear estimator $\boldsymbol{b}_0$, wlog assume it is $\boldsymbol{b}_0 = \boldsymbol{C}\boldsymbol{y}$ for some matrix $\boldsymbol{C}$.

For any $\boldsymbol{C}$ we can always find a matrix $\boldsymbol{D} = \boldsymbol{C} - \boldsymbol{A}$ so that t

$$\boldsymbol{b}_0 \;=\; \boldsymbol{C}\boldsymbol{y} \;=\; (\boldsymbol{D} + \boldsymbol{A})\boldsymbol{y}$$

By (A1) $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \varepsilon$, therefore

$$\begin{aligned}\boldsymbol{b}_0 =& (\boldsymbol{D} + \boldsymbol{A})\boldsymbol{y}\\ =& \boldsymbol{D}\boldsymbol{y} + \boldsymbol{A}\boldsymbol{y}\\ =& \boldsymbol{D}(\boldsymbol{X}\boldsymbol{\beta} + \varepsilon) + \boldsymbol{b}\\ =& \boldsymbol{D}\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{D}\varepsilon + \boldsymbol{b}\end{aligned}$$

By assumption $\boldsymbol{b}_0$ it is unbiased conditional on $\boldsymbol{X}$, therefore $\mathrm{E}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] = \boldsymbol{\beta}$.

Notes

Therefore

$$\begin{aligned}\mathrm{E}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] =& \mathrm{E}[\; \boldsymbol{D}\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{D}\varepsilon + \boldsymbol{b} \;\,|\, \boldsymbol{X}]\\ =& \boldsymbol{D}\boldsymbol{X}\boldsymbol{\beta} + \mathrm{E}[\boldsymbol{D}\varepsilon \,|\, \boldsymbol{X}] + \mathrm{E}[\boldsymbol{b} \,|\, \boldsymbol{X}]\\ =& \boldsymbol{D}\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{D}\,\mathrm{E}[\varepsilon \,|\, \boldsymbol{X}] + \boldsymbol{\beta}\\ =& \boldsymbol{D}\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\beta}\end{aligned}$$

The fact that $\boldsymbol{b}_0$ is unbiased implies $\boldsymbol{D}\boldsymbol{X} = \boldsymbol{0}$, hence

$$\boldsymbol{b}_0 = \boldsymbol{D}\varepsilon + \boldsymbol{b} \tag{5.1}$$

Now substract $\boldsymbol{\beta}$ from both side of (5.1) and obtain the estimation error $\boldsymbol{b}_0$

$$\boldsymbol{b}_0 - \boldsymbol{\beta} \;=\; \boldsymbol{D}\varepsilon + (\boldsymbol{b} - \boldsymbol{\beta}) \;=\; (\boldsymbol{D} + \boldsymbol{A})\varepsilon$$

Notes

In order to show that the OLS is BLUE we need to show that for any $\boldsymbol{b}_0$ (linear in $\boldsymbol{y}$ and unbiased) it holds true that $\mathrm{Var}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}]$ is bigger then $\mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}]$

We need to show that $\mathrm{Var}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] \succeq \mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}]$, that is

$$\mathrm{Var}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] - \mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}] \quad \text{is PSD.}$$

Same trick as before: the variance of a constant is zero, so we start from $\mathrm{Var}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] = \mathrm{Var}[(\boldsymbol{b}_0 - \boldsymbol{\beta}) \,|\, \boldsymbol{X}]$.

Notes

$$\begin{aligned}
\mathrm{Var}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] &= \mathrm{Var}[(\boldsymbol{b}_0 - \boldsymbol{\beta}) \,|\, \boldsymbol{X}] \\
&= \mathrm{Var}[(\boldsymbol{D} + \boldsymbol{A})\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}] \\
&= (\boldsymbol{D} + \boldsymbol{A})\mathrm{Var}[\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}](\boldsymbol{D} + \boldsymbol{A})' \\
&= \sigma^2(\boldsymbol{D} + \boldsymbol{A})(\boldsymbol{D}' + \boldsymbol{A}') \\
&= \sigma^2(\boldsymbol{D}\boldsymbol{D}' + \boldsymbol{A}\boldsymbol{D}' + \boldsymbol{D}\boldsymbol{A}' + \boldsymbol{A}\boldsymbol{A}')
\end{aligned}$$

Now

$$\boldsymbol{D}\boldsymbol{A}' = \boldsymbol{D}\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} = \boldsymbol{0}$$

$$\boldsymbol{A}\boldsymbol{A}' = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} = (\boldsymbol{X}'\boldsymbol{X})^{-1}$$

$$\mathrm{Var}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] = \sigma^2[\boldsymbol{D}\boldsymbol{D}' + (\boldsymbol{X}'\boldsymbol{X})^{-1}] = \sigma^2\boldsymbol{D}\boldsymbol{D}' + \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}$$

Notes

Observe that

- any covariance matrix must be PSD
- the sum of PSD matrices is PSD

Since

$$\mathrm{Var}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] = \sigma^2 \boldsymbol{D}\boldsymbol{D}' + \mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}]$$

$\sigma^2 \boldsymbol{D}\boldsymbol{D}'$ must be PSD

But

$$\sigma^2 \boldsymbol{D}\boldsymbol{D}' = \mathrm{Var}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] - \mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}]$$

which implies that the right-hand-side must be PSD. Hence

$$\mathrm{Var}[\boldsymbol{b}_0 \,|\, \boldsymbol{X}] \succeq \mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}]$$

which completes the proof ■

Notes

Proof of part (c). First note that $\boldsymbol{e} = \boldsymbol{M}\boldsymbol{\varepsilon}$, in fact recall that $\boldsymbol{M}\boldsymbol{X} = \boldsymbol{0}$ and

$$\boldsymbol{e} = \boldsymbol{M}\boldsymbol{y} = \boldsymbol{M}(\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \boldsymbol{M}\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{M}\boldsymbol{\varepsilon} = \boldsymbol{M}\boldsymbol{\varepsilon}$$

Now work out the formula of covariance between vectors:

$$\begin{aligned}
\mathrm{Cov}[\boldsymbol{b}, \boldsymbol{e} \,|\, \boldsymbol{X}] &= \mathrm{E}[(\boldsymbol{b} - \mathrm{E}[\boldsymbol{b}|\boldsymbol{X}])(\boldsymbol{e} - \mathrm{E}[\boldsymbol{e}|\boldsymbol{X}])' \,|\, \boldsymbol{X}] \\
&= \mathrm{E}[(\boldsymbol{b} - \boldsymbol{\beta})(\boldsymbol{e})' \,|\, \boldsymbol{X}] \\
&= \mathrm{E}[(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\varepsilon}\,(\boldsymbol{M}\boldsymbol{\varepsilon})' \,|\, \boldsymbol{X}] \\
&= \mathrm{E}[(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\varepsilon}\,\boldsymbol{\varepsilon}'\boldsymbol{M} \,|\, \boldsymbol{X}] \\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\,\mathrm{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \,|\, \boldsymbol{X}]\boldsymbol{M} \quad \text{(pull out what's known)} \\
&= \sigma^2 (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{M}
\end{aligned}$$

Since $\boldsymbol{X}'\boldsymbol{M} = (\boldsymbol{M}\boldsymbol{X})' = \boldsymbol{0}$, then $\mathrm{Cov}[\boldsymbol{b}, \boldsymbol{e} \,|\, \boldsymbol{X}] = \boldsymbol{0}$ ■

Notes

Finite sample properties of s^2

In order to do inference on $\boldsymbol{\beta}$, we need the variance of $\boldsymbol{b}$, but it depends on σ^2

Since ε_i is recovered based on e_i, the natural analog estimator of σ^2 would be

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} e_i^2 = \frac{\boldsymbol{e}'\boldsymbol{e}}{n}$$

However, this is biased because residuals are a two stages estimates of ε_i. In fact, we need to pre-estimate $\boldsymbol{\beta}$ to get $\boldsymbol{e}$.

An unbiased estimator is obtained by correcting the biased one with the degrees of freedom as usual, that is

$$s^2 = \frac{1}{n-K}\sum_{i=1}^{n} e_i^2 = \frac{\boldsymbol{e}'\boldsymbol{e}}{n-K}$$

Notes

Proposition: unbiased estimation of σ^2

Assume (A1), (A2), (A3), (A4), then s^2 is an unbiased estimator of σ^2.

Proof.

$$\begin{aligned}
\mathrm{E}[\boldsymbol{e}'\boldsymbol{e} \mid \boldsymbol{X}] &= \mathrm{E}[(\boldsymbol{M}\boldsymbol{\varepsilon})'\boldsymbol{M}\boldsymbol{\varepsilon} \mid \boldsymbol{X}] \\
&= \mathrm{E}[\boldsymbol{\varepsilon}'\boldsymbol{M}'\boldsymbol{M}\boldsymbol{\varepsilon} \mid \boldsymbol{X}] \\
&= \mathrm{E}[\boldsymbol{\varepsilon}'\boldsymbol{M}\boldsymbol{\varepsilon} \mid \boldsymbol{X}] \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n} m_{ij}\,\mathrm{E}[\varepsilon_i\varepsilon_j \mid \boldsymbol{X}] \quad \text{(verify this!)} \\
&= \sigma^2 \sum_{i=1}^{n} m_{ii} \\
&= \sigma^2\,\mathrm{tr}(\boldsymbol{M})
\end{aligned}$$

Notes

$$\mathrm{tr}(\boldsymbol{M}) = \mathrm{tr}(\boldsymbol{I}_n - \boldsymbol{P}) = n - \mathrm{tr}(\boldsymbol{P})$$

and

$$\mathrm{tr}(\boldsymbol{P}) = \mathrm{tr}\left(\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\right) = \mathrm{tr}\left(\boldsymbol{X}'\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\right) = \mathrm{tr}(\boldsymbol{I}_k) = K$$

Therefore $\mathrm{E}[\boldsymbol{e}'\boldsymbol{e} \,|\, \boldsymbol{X}] = (n-K)\sigma^2$. And this proves that the analog estimator $\hat{\sigma}^2$ is biased, in fact

$$\mathrm{E}\left[\hat{\sigma}^2 \,|\, \boldsymbol{X}\right] = \mathrm{E}\left[\frac{\boldsymbol{e}'\boldsymbol{e}}{n} \,\middle|\, \boldsymbol{X}\right] = \frac{n-K}{n}\sigma^2$$

This also shows that s^2 is unbiased, in fact

$$\mathrm{E}[s^2 \,|\, \boldsymbol{X}] = \mathrm{E}\left[\frac{\boldsymbol{e}'\boldsymbol{e}}{n-K} \,\middle|\, \boldsymbol{X}\right] = \frac{n-K}{n-K}\sigma^2 = \sigma^2$$

■

Notes

Estimated standard errors of $\boldsymbol{b}$

We know that

$$\mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}] = \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}$$

which depends on population quantities.

We can estimate the variance matrix of $\boldsymbol{b}$ based on the plug-in principle:

$$\widehat{\mathrm{Var}\,(\boldsymbol{b})} = s^2(\boldsymbol{X}'\boldsymbol{X})^{-1}$$

Let $\mathrm{Se}(b_k)$ be the square root of kth element on the diagonal of $\widehat{\mathrm{Var}\,(\boldsymbol{b})}$, then $\mathrm{Se}(b_k)$ is the standard error of the b_k parameter estimate, that is

$$\mathrm{Se}(b_k) = s\sqrt{(\boldsymbol{X}'\boldsymbol{X})^{-1}_{kk}}$$

Notes